

Cello Quantitative Research Plan

Cello Capital Management

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Vision Statement

- Build a comprehensive software framework (Scale) for the pricing, hedging and risk management of fixed-income instruments and, in particular, structured products arising in the housing and consumer finance markets. The library will be written in C++.
- Build a database (ScaleDB) that maintains proprietary loan- and pool-level indicators of prepayment risk.

Benefits of Scale and ScaleDB

- Micro-level specification of proprietary models
- Reduce low-level model building work and allow focus on more complex problems
- Platform for scaling the firm across asset classes and strategies
- Tangible repository for Cello IP
- Training ground for new Associates/Analysts in the Quantitative Research group

Scale Accomplishments: 2016

- Yield Curve Builder (Libor/Swap Curve)
- 1-factor Hull-White Interest Rate Model
- 30-year Conventional Mortgage Rate Model
- Pass-through and IOS cash flow engine
- 30-year Fannie Mae Prepayment model
- Conventional House Price model
- Risk Engine (OAS, Durations etc.)
- Roll-rate Model for GSE Credit Risk Transfer Deals (only implemented in R)
- Utility functions: Dates, Splines, Optimizer/Solver etc.
- Integration with Quantlib

- Quantlib is a free/open-source library for modeling, trading and risk management
- Reasonably well-tested: It's been around since 2000
- Well-maintained: There are at least a couple of releases every year
- Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
- Documented code

Scale: Software Engineering Processes

- Documentation
- Source control and software versioning
- One-step daily builds
- Bug database
- Unit testing/Error checking
- Coding standards
- Logging

ScaleDB Accomplishments: 2016

- Mortgage rate series for non-traditional loan types (Jumbo conforming, FHA loans)
- Likelihood that a loan was HARPed
- Likelihood that a loan is HARP-eligible
- Multiple versions of the Media Effect
- Multiple version of Mortgage Credit Availability Index
- Refinance incentive calculations that take LLPAs, PMI, and the ability to refinance across different programs into account
- Burnout

- Miscellaneous improvements to 30-year Fannie Mae prepayment model (Q1)
- Roll out 30-year Ginnie Mae prepayment model (Q2) and 15-year Fannie Mae model (Q3)
- Ginnie Mae Project Loan Prepayment Model and Pricing Analytics (Q1-Q3)
- Improvement to Interest Rate Model: stability of calibration (Q1), incorporation of more yield curve factors (Q2-Q4)